



Fatigue (PDQ®)–Health Professional Version

Overview

Cancer-related fatigue (CRF) is a distressing, persistent, subjective sense of physical, emotional, and/or cognitive tiredness or exhaustion related to cancer or cancer treatment that is not proportional to recent activity and interferes with usual functioning.[1] Fatigue is the most common side effect of cancer treatment with chemotherapy, radiation therapy, bone marrow transplantation, or selected biologic response modifiers.[2] Clinically significant levels of fatigue may also negatively impact survival.[3][[Level of evidence: III](#)] The specific mechanisms underlying a common pathophysiology for CRF are unknown.

Cancer treatment–related fatigue is a commonly reported symptom, with 80% of patients reporting fatigue while receiving chemotherapy or radiation therapy.[4] The condition generally improves after therapy is completed, but some level of fatigue may persist for months or years after treatment. For a subset of patients, fatigue may be a significant issue long into survivorship.[5,6] For example, a longitudinal study assessed fatigue in individuals with stage I to stage III breast cancer over three time points postdiagnosis (1 year, n = 5,640; 2 years, n = 5,000; 4 years, n = 3,400). The study found that over 30% of patients at each time point experienced severe global fatigue.[7][[Level of evidence: III](#)] Physical fatigue (35%) occurred more often than emotional fatigue (25%) or cognitive fatigue (13%).[8] Fatigue is also seen as a presenting symptom in cancers that cause complications such as anemia, endocrine dysfunction, neuromuscular complications, psychological distress, and end-organ dysfunction (e.g., renal, pulmonary, or cardiac dysfunction). Fatigue is common in people with advanced cancer who are not undergoing active cancer treatment. Cancer treatment-related fatigue has been reported in 39% to more than 90% of patients undergoing cancer treatment [9-13] and in 19% to 82% of patients posttreatment.[4,14]

Fatigue experienced as a side effect of cancer treatment is differentiated from fatigue experienced by healthy people in their daily lives. **Healthy fatigue** is frequently described as acute fatigue that is eventually relieved by sleep and rest; **cancer treatment-related fatigue** is categorized as chronic fatigue because it is present over a long period of time, interferes with functioning, and is not completely relieved by sleep and rest.[15] Also, the level of CRF is often disproportionate to the level of activity or energy exerted.[15] Although the label **chronic fatigue** is accurate, it does not mean that people with cancer who experience fatigue have chronic fatigue syndrome. Using this phrase can be confusing to both patients and health professionals. Terms such as **cancer fatigue**, **cancer-related fatigue**, and **cancer treatment-related fatigue** have all been used in the clinical literature, research literature, and educational materials for patients and the public.

Fatigue, like pain, is a self-perceived state and patient-reported outcome. Patients may describe fatigue as feeling:[16]

- Tired.
- Weak.
- Exhausted.
- Lazy.
- Weary.
- Worn-out.
- Heavy.
- Slow.

Questions?

- Like they have no energy or get-up-and-go.

Health professionals have included fatigue within concepts such as:

- Asthenia.
- Lassitude.
- Malaise.
- Prostration.
- Exercise intolerance.
- Lack of energy.
- Weakness.

Studies of women with breast cancer have attempted to define specific fatigue trajectories. For example, some patients experience a high degree of fatigue during treatment and recovery, while others deteriorate over time. In contrast, some patients suffer from little fatigue throughout treatment. Suggested fatigue trajectories include the following:[8,17]

- Very low fatigue.
- Low fatigue.
- Late or deteriorating fatigue (initially low symptoms that increase over time).
- Recovery (initially high symptoms that decrease over time).
- High fatigue.

Research on fatigue in people with cancer has included primarily self-reports of fatigue, with increasing data exploring biological or physiological correlates. Such correlates have included measures of muscle weakness, maximal oxygen uptake, cytokines, cortisol, and genetic biomarkers.[10]

Fatigue has a negative impact on all areas of function, including the following:[18-21]

- Mood.
- Physical function.
- Work performance.
- Social interaction.
- Family care.[22]
- Cognitive performance.
- School work.
- Community activities.
- Sense of self.
- Activities of daily living in older cancer survivors.[23]

The pattern of fatigue associated with cancer treatment varies according to the type and schedule of treatment. For example, people treated with cyclic chemotherapy regimens generally exhibit peak fatigue in the days following treatment, then lower levels of fatigue until the next treatment. However, patients undergoing external-beam radiation therapy report gradually increasing fatigue over the course of therapy of the largest treatment field. Few studies of people undergoing cancer treatment have addressed the issue of fatigue as a result of the emotional distress associated with undergoing a diagnostic evaluation for cancer and the effects of medical and surgical procedures used for evaluation and for initial treatment. Because most adults enter the cancer care system following at least one surgical

procedure, and because surgery and emotional distress are both associated with fatigue, it is likely that most people beginning nonsurgical treatment are experiencing fatigue at the beginning of treatment.[21,24]

Fatigue management focuses on identifying and treating the underlying factors that may be contributing to fatigue. Most clinical recommendations for managing the symptoms of fatigue caused by something other than chemotherapy-induced anemia rely on careful development of clinical hypotheses, as outlined in the National Comprehensive Cancer Network (NCCN) guidelines on fatigue.[1] NCCN category 1 interventions for CRF include the following:

- Physical activities (e.g., yoga).
- Massage therapy.
- Psychosocial interventions (e.g., cognitive behavioral therapy/behavioral therapy, supportive expressive therapies, and psychoeducational therapies).

For more information, see the [Interventions](#) section.

Although much progress has been made, further research is needed to better define fatigue and its trajectory, understand its physiology, and determine the best ways to prevent and treat it.

In this summary, unless otherwise stated, evidence and practice issues as they relate to adults are discussed. The evidence and application to practice related to children may differ significantly from information related to adults. When specific information about the care of children is available, it is summarized under its own heading.

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Pathogenesis of Fatigue

Except for chemotherapy-induced anemia, the mechanisms responsible for fatigue in people with cancer are not known. Understanding the causes of fatigue in people with cancer is especially challenging because each individual may experience multiple possible causes of fatigue simultaneously. Multiple underlying etiological factors beyond the type and treatment of cancer have been proposed, including psychological distress, life demands, sleep disturbance, neurophysiological changes, disruption of circadian rhythms, cardiac issues, neuroimmunological changes, and genetic variations.[1]

Growing evidence, particularly for women with breast cancer and men with prostate cancer, suggests that fatigue is associated with markers of increased immune inflammatory activity. When fatigued individuals with a history of breast cancer are compared with breast cancer survivors without fatigue, different patterns emerge with respect to interleukin-6, interleukin-1 receptor antagonist, C-reactive protein, neopterin, and soluble tumor necrosis factor receptor-II.[2-5] Although the precise relationships—and the clinical meaning of those relationships—are not yet known, increased cytokines likely contribute to the symptoms of asthenia, fatigue, and lethargy. However, so far no large, well-controlled studies have evaluated the effects of general anti-inflammatory agents on fatigue or cytokine biomarkers.

Other studies demonstrate a change in the regulation of cortisol by the hypothalamic pituitary adrenal axis. One key study put fatigued and nonfatigued breast cancer survivors through a stress battery in a laboratory setting. Nonfatigued survivors mounted a significant cortisol increase in response to acute stress, while fatigued survivors had a very blunted response.[6] Another study has shown that fatigued breast cancer survivors have flattened cortisol slopes, having higher levels of cortisol at the end of the day than do nonfatigued survivors.[7] It is the dysregulation of the hypothalamic-pituitary-adrenal (HPA) axis that may account for the prolonged inflammatory cytokine milieu. Understanding the body's response to numerous chronic stressors in cancer may help in managing fatigue.

Finally, another theory is that chronic exposure to proinflammatory cytokines negatively impacts serotonin levels. One hypothesis is that the relationship between central nervous system concentrations of serotonin and fatigue have a U-shaped relationship, suggesting that very high and very low levels of serotonin may be associated with cancer-related fatigue.[8] However, studies that have evaluated serotonergic agents have not demonstrated a benefit for fatigue.[9] The role and relationship of many important neurotransmitters such as dopamine, norepinephrine, and serotonin with HPA axis functioning and cytokine expression have yet to be fully understood.

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Contributing Factors

Although fatigue is clearly prevalent in patients with cancer, it has been difficult to identify consistent correlates of fatigue in this population. The factors most often implicated include the following:[1,2]

- Cancer treatment.[3]
- Anemia.
- Medications.
- Hormonal therapy.[4]
- Cachexia/anorexia.

- Metabolic disturbances.
- Hormone deficiency or excess.
- Psychological distress, including depression.[5]
- Physical deconditioning.[6]
- Sleep disturbances.[3,7]
- Excessive inactivity.
- Tobacco use.[4]
- High body mass index (BMI).[4]
- Neuromuscular dysfunction.
- Pain and other symptoms.
- Proinflammatory cytokines.
- Nutritional deficiencies.
- Dehydration.
- Infection.
- Concomitant medical illness.
- Pretreatment fatigue.[8]

Fatigue has also been identified as a common symptom in symptom clusters associated with cancer, e.g., fatigue-pain and fatigue-cognitive clusters. For more information, see [Symptom Clusters in Cancer](#).

Cancer Treatment

The association of fatigue with the major cancer treatment modalities of surgery, chemotherapy, radiation therapy, endocrine therapy, and biologic response modifier therapy caused speculation that fatigue resulted from tissue damage or accumulation of the products of cell death. Interest in the effects of cancer treatment on the production of proinflammatory cytokines is based on recognition of the strong fatigue-inducing effect of some biologic response modifiers such as interferon-alpha and the finding of elevated levels of proinflammatory cytokines in people experiencing persistent fatigue after cancer treatment.[9,10] In longitudinal studies of patients undergoing radiation therapy, polymorphisms in tumor necrosis factor-alpha and interleukin-6 were associated with elevated fatigue before, during, and for 4 months after completion of treatment.[11,12]

Many people with cancer undergo surgery for diagnosis or treatment. Despite the high incidence of postoperative fatigue observed in these patients in clinical practice, few investigators have examined its causes and correlates. It is clear, however, that fatigue postsurgery improves with time and is compounded by fatigue caused by other cancer treatments.[13,14]

Fatigue has long been associated with radiation exposure and is reportedly one of the most common and activity-limiting side effects of radiation therapy for cancer.[15,16] Up to 90% of patients undergoing radiation therapy experience fatigue during the course of their treatment.[17] Most of the research describing the fatigue trajectory during radiation therapy has been conducted in women with breast cancer and men with prostate cancer.[3,18]

Fatigue increases throughout radiation therapy, peaking around mid-course and remaining at this level until radiation therapy is completed. It then improves somewhat during the 2 months after treatment ends. A study that investigated the trajectory of fatigue in men who were undergoing radiation therapy for prostate cancer (N = 82) found significant interindividual variability.[3] The authors used hierarchical linear modeling, a highly sophisticated analytical method, to identify predictors for prolonged fatigue trajectories. Younger men with high levels of fatigue at the initiation of radiation therapy were at increased risk of developing higher levels of morning and evening fatigue during radiation

therapy. In addition, the level of depression at the initiation of radiation therapy predicted the level of morning fatigue during radiation therapy.[3]

A second study of 73 women who were undergoing adjuvant radiation therapy for breast cancer found similar differences in the patterns and predictors of morning versus evening fatigue.[7] Participants were recruited to the study at their simulation visits and completed baseline questionnaires. Data were then collected on 2 subsequent days, in the morning and at bedtime, each week during radiation therapy; every 2 weeks for 2 months after radiation therapy; and once a month for 2 additional months thereafter. Fatigue was measured with the Lee Fatigue Scale. For the group as a whole, over the 25 weeks of data collection, morning fatigue decreased slightly during radiation therapy and was constant for 4 months afterwards, while evening fatigue increased through radiation therapy and then declined slightly after treatment. Evening fatigue was higher for those who:

- Were working.
- Had children at home.
- Had higher depression scores.

Morning fatigue was higher for those who:

- Had more trait anxiety.
- Were experiencing sleep disturbance.
- Were younger.
- Had lower body mass indices.

Advanced disease and comorbidities also added to the severity of morning fatigue.[7][[Level of evidence: III](#)]

Several research studies document a fatigue syndrome that is not specific to disease type or radiation site and that demonstrates a gradual decline in fatigue in the patient after treatment is completed.[19,20] Some of these studies suggest, however, that not all patients return to pretreatment energy levels. Risk factors for persistent low energy in cancer patients include pretreatment fatigue, psychological distress, high BMI, tumor location, advanced disease, and combination-modality therapy.[8,14,21]

Fatigue is a dose-limiting toxicity of treatment with a variety of biotherapeutic agents. Biotherapy exposes patients with cancer to exogenous and endogenous cytokines.[22] Biotherapy-related fatigue usually occurs as part of a constellation of symptoms called *flu-like syndrome*.

Mental fatigue and cognitive deficits have also been identified as biotherapy side effects. The type of biotherapeutic agent used may influence the type and pattern of fatigue.[23,24]

Treatment with chemotherapy is a predictor of fatigue and can be exacerbated by the coexistence of pain, depression, and/or anxiety.[25][[Level of evidence: II](#)]; [26] A longitudinal, descriptive study reported the highest levels of fatigue at the midpoint of a patient's chemotherapy cycles, with fatigue improving after treatment but not quite returning to baseline levels 30 days after the last treatment.[25] In another longitudinal study of women with stage 0 to stage II breast cancer who received chemotherapy with or without radiation therapy (n = 103) versus radiation therapy alone (n = 102) versus a control group (n = 193),[27] increases in fatigue were demonstrated 3 years posttreatment for the group that received chemotherapy with or without radiation therapy, compared with the two other groups. Mean scores for fatigue severity measured by the Fatigue Symptom Inventory (range, 0–10) increased over the 3 years.

A longitudinal, descriptive study of 78 women with gynecological cancer examined the daily and intraday changes and interrelationships among fatigue, depression, and disruptions in sleep and activity, before and after each individual chemotherapy treatment, for three treatments. Significant changes in symptoms were noted over time. Before infusions, fatigue was associated with depression; after infusions, fatigue was significantly associated with increased

depression and sleep/wake irregularities (increases in minutes awake at night and decreases in daytime activity and sleep/wake activity).[28]

Aromatase inhibitors—the recommended first-line adjuvant endocrine therapy in postmenopausal women with hormone receptor–positive breast cancer—have been linked to cancer-related fatigue (CRF). In one study of survivors of stage 0 to III breast cancer who were receiving adjuvant aromatase inhibitor therapy at an outpatient breast oncology clinic, 616 of 1,103 participants (55.8%) had moderate to severe CRF.[29] In addition, breast cancer survivors who were younger (age ≤55 years), were college educated, had higher body mass indices, and reported more pain and insomnia were more likely to have moderate to severe CRF than were their counterparts.

Treatment with immune checkpoint inhibitors has been associated with clinically significant fatigue. Reviews of outcomes data demonstrated that fatigue is the most common adverse event incorporating anti–programmed death 1 (PD-1)/programmed death-ligand 1 (PD-L1) agents. In the first phase I studies of nivolumab, 16% to 24% of patients had treatment-related fatigue, and 1% to 2% had grade 3 or 4 severity. Single-agent immune checkpoint studies have reported an incidence of 16% to 37% with anti-PD-1 agents, and 12% to 24% with anti-PD-L1 agents. Clinical studies that combined anti-PD-1/PD-L1 agents with other immune checkpoint inhibitors reported even higher rates of fatigue, in up to 71% of patients. The specific mechanism by which immune checkpoint inhibitors cause fatigue is not known; however, when fatigue symptoms occur during treatment, clinicians should be vigilant in assessing for early symptoms of endocrine dysfunction, such as hypothyroidism.[30-33]

Anemia

Evidence suggests that anemia may be a major factor in CRF and quality of life in cancer patients.[34-36] Anemia can be related to the disease itself or caused by therapy. Occasionally, anemia is simply a co-occurring medical finding that is related to neither disease nor therapy. Anemia is often a significant contributor to symptoms in people with cancer. For individual patients, it can be difficult to discern the actual impact of anemia because there are often other problems that confound the ability to weigh the specific impact of anemia.[37]

A retrospective review was conducted to understand anemia in patients undergoing radiation therapy. Anemia was found in 48% of the patients initially and increased to 57% during therapy. It was more common in women than in men (64% vs. 51%); however, men with prostate cancer experienced the greatest increase in anemia during radiation therapy.[38] In certain cancers, such as cancer of the cervix and cancer of the head and neck, anemia has been found to predict poor survival and diminished quality of life in patients undergoing radiation therapy.[39-41]

Nutrition Factors

Fatigue often occurs when the body's energy requirements exceed the supply of energy sources. In people with cancer, three major mechanisms may be involved:

- Alteration in the body's ability to process nutrients efficiently.
- Increase in the body's energy requirements.
- Decrease in intake of energy sources.

Causes of nutritional alterations are listed in Table 1.

Table 1. Nutrition/Energy Factors in Cancer

Mechanisms	Causes
Altered ability to process nutrients	Impaired glucose, lipid, and protein metabolism

Mechanisms	Causes
Increased energy requirements	Tumor consumption of and competition for nutrients
	Hypermetabolic state due to tumor growth
	Infection/fever
	Dyspnea
Decreased intake of energy sources	Anorexia
	Nausea/vomiting
	Diarrhea
	Bowel obstruction

A randomized controlled trial compared a plant-based, high-protein diet to usual care in 103 patients with newly diagnosed breast cancer who were undergoing adjuvant chemotherapy.[42][[Level of evidence: I](#)] Patients were assessed over three time points (T0: baseline, T1: end of third chemotherapy infusion, T2: 3 weeks after last treatment). Fatigue was measured using the Fatigue Symptom Inventory, which has a clinically meaningful fatigue level threshold of more than 3. Patients in the control group (n = 51) had an increase in mean fatigue score from 4.2 + 1.64 to 5.37 + 1.87, while patients in the intervention group (n = 52) had a decrease in mean fatigue score from 4.2 + 1.94 to 2.47 + 1.31 ($P < .001$). Of note, the intervention group's fatigue level decreased to below a clinically meaningful level between T0 and T2, while the control group's fatigue level stayed above the clinically meaningful threshold and slightly worsened. In addition, while both groups had a decrease in BMI (intervention group, 0.7 + 0.8 kg/m²; control group, 0.4 + 1.3 kg/m²), the control group had a decrease in muscle mass, and the intervention group had an increase in muscle mass and a decrease in fat mass ($P < .001$). A plant-based, high protein diet may be an effective approach to fatigue management.

Psychological Factors

Numerous factors related to the moods, beliefs, attitudes, and reactions to stressors of people with cancer can also contribute to the development of chronic fatigue. Anxiety and depression are the most common comorbid psychiatric disorders of CRF.[43][[Level of evidence: II](#)] Often, fatigue is the final common pathway for a range of physical and emotional etiologies.

Depression can be a comorbid, disabling syndrome that affects approximately 15% to 25% of people with cancer.[28,44] Multiple studies show that pretreatment depression increases fatigue during and after cancer treatment.[9,45,46] The presence of depression—manifested by loss of interest in normal activities, difficulty concentrating, lethargy, and feelings of hopelessness—can compound the physical causes of fatigue in these individuals and persist long past the time when physical causes have resolved.[47] A history of stressful experiences in childhood, including abuse and neglect, has also been associated with higher levels of fatigue in breast cancer survivors.[48]

The anxiety and fear associated with a cancer diagnosis—and the impact of that diagnosis on a person's physical, psychosocial, and financial well-being—are sources of emotional stress. Distress associated with the cancer diagnosis alone may trigger fatigue. A study of 74 early-stage breast cancer patients with no history of affective disorder assessed

various symptoms of adjustment approximately 2 weeks after diagnosis. About 45% of participants noted moderate or high levels of fatigue. This fatigue may have been secondary to the increased cognitive strain of dealing with the diagnosis or to insomnia, reported as moderate to severe by about 60% of patients. Therefore, fatigue may begin before treatment as a result of worry or other cognitive factors, both primary and secondary to insomnia. Various forms of treatment may compound this fatigue.[49]

In cancer survivors, fatigue may also be above levels seen in the general population.[50,51] A Brazilian study found that in patients who had advanced cancer but were not undergoing therapy, those with anxiety and depression had higher fatigue levels.[51] A Dutch study found a correlation between anxiety, depression, and CRF.[52] Despite psychological care, those with poorer physical health and mood disturbances reported more fatigue. For more information, see [Depression](#) and [Adjustment to Cancer: Anxiety and Distress](#).

Psychological and symptom distress have also been found to be significant predictors of fatigue.[53,54] In a study of 101 women about to undergo surgery for breast cancer, younger age, presurgery distress, and expectations about fatigue significantly predicted fatigue levels 1 week after surgery. In the regression model, age, distress, and expectancy each uniquely contributed to fatigue, with distress and expectancy accounting for 25% of the variance.[53][[Level of evidence: III](#)] In a longitudinal study with women who had gynecological cancer, symptom and psychological distress significantly predicted fatigue before, during, and after treatment with chemotherapy, explaining up to 80% of the variance in fatigue scores after chemotherapy treatment.[54] In another study, posttreatment colorectal cancer patients were found to have more fatigue when they had catastrophizing thoughts (rumination, magnification, and helplessness).[55] Factors similar to those seen in patients with early-stage cancer also contribute to fatigue in patients with advanced, incurable cancer.[56]

Cognitive Factors

Impairment in cognitive functioning, including decreased attention span and impaired perception and thinking, is commonly associated with fatigue.[57] Although fatigue and cognitive impairments are linked, the mechanism underlying this association is unclear. The mental demands inherent in the diagnosis and treatment of cancer have been well documented, but little is known about the concomitant problem of *attention fatigue* in people with cancer. Attention problems are common during and after cancer treatment.[57] Some of these problems may be caused by the fatigue of *directed attention*.[\[57,58\]](#) Attention fatigue may be relieved by activities that promote rest and restore directed attention.[\[58\]](#) Although sleep is necessary for relieving attention fatigue and restoring attention, it is insufficient when attention demands are high. Research in this area is limited and most commonly conducted in breast cancer patients, potentially limiting its application across diverse populations. Empirical literature suggests that exposure to the natural environment may help restore directed attention and relieve attention fatigue.

Sleep Disorders and Inactivity

Causative or contributing factors in CRF may be:

- Disrupted sleep.
- Poor sleep hygiene.
- Decreased nighttime sleep or excessive daytime sleep.
- Inactivity.
- Timing of therapy.[\[28\]](#)

Patients with less daytime activity, restless sleep, and obesity were noted to have consistently higher levels of CRF.[\[59\]](#)

Sleep disorders clearly contribute to fatigue [\[60\]](#) and may differentially affect fatigue ratings, depending on the time of the rating. A study that evaluated fatigue in women undergoing radiation therapy for breast cancer found that sleep had a greater influence on fatigue values in the morning than in the evening.[\[7\]](#) However, fatigue and sleep can also be

distinct problems. One study found that the use of cognitive behavioral therapy resulted in significant improvement in sleep quality but did not significantly affect CRF.[61] For more information, see [Sleep Disorders](#).

Other Medications That Contribute to Fatigue

Medications other than chemotherapy drugs may contribute to fatigue. Opioids used to treat cancer-related pain are often associated with sedation, though the degree varies among individuals. Opioids are known to alter the normal function of the hypothalamic secretion of gonadotropin-releasing hormone.[62] Hypogonadism may be found in patients with advanced cancer and can contribute to fatigue during cancer treatment.[63,64] One case-control study examined the effects of chronic oral opioid administration in survivors of cancer and, consistent with the research on intrathecal administration, found marked central hypogonadism among the opioid users with significant symptoms of sexual dysfunction, depression, and fatigue.[65] In patients with hypogonadism and symptoms of fatigue, testosterone replacement over a month had mixed results in clinical trials, with benefit for fatigue occurring around the 70-day mark, but with no improvement in quality of life.[66]

Other medications—including tricyclic antidepressants, neuroleptics, beta blockers, benzodiazepines, and antihistamines—may produce side effects of sedation. In addition, concurrent medications such as analgesics, hypnotics, antidepressants, antiemetics, steroids, or anticonvulsants—many of which act on the central nervous system—can significantly compound the problem of fatigue. The coadministration of multiple drugs with varying side effects may compound fatigue symptoms.

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Assessment

The first step in the assessment of fatigue is screening.[1] Patients can be screened for fatigue at the initial visit, at the beginning and end of primary cancer treatments, and at least annually (or as clinically indicated) during follow-up care. Evidence indicates that brief, self-report, quantitative, and single-item assessments with empirically established cut-off scores can measure fatigue levels in an expedited manner.[2] These tools include assessments such as the National Comprehensive Cancer Network (NCCN) intensity tool [2] and the visual analog scale (VAS),[3] which are 0-to-10 numeric rating scales (0 = no fatigue; 10 = worst fatigue imaginable). Ratings are categorized as none to mild (score, 0–3), moderate (score, 4–6), and severe (score, 7–10). Fatigue is considered clinically significant when rated in the moderate-to-severe range (score, 4–10).[4]

Patients with moderate-to-severe levels of fatigue require further evaluation. One study of ambulatory outpatients with solid tumors (n = 148) evaluated the usefulness of single-item screening for symptoms such as fatigue and pain.[5] Investigators found that the single-item assessment can help identify patients who require comprehensive assessments of their symptoms. Patients identified through single-item screening tools undergo comprehensive assessments to detect clinically relevant symptomatology.[5,6]

Cancer-related fatigue (CRF) is multifactorial. The purpose of an in-depth evaluation is to assess diverse factors that can cause or contribute to fatigue.[7-10] Such an evaluation may identify factors that can be reversed or treated (e.g., hypothyroidism, sleep disturbances, or depression). In addition, fatigue can be part of a symptom cluster. For more information, see [Symptom Clusters in Cancer](#).

A comprehensive assessment of a fatigued patient starts with carefully obtaining a history to fully characterize the patient's fatigue pattern and to identify all factors that contribute to its development. An in-depth evaluation of fatigue includes the following:

- Status of cancer and cancer treatments: recurrence or progression of disease, type and length of cancer treatments, and capacity of treatments to induce fatigue.
- Review of systems to assess impact of cancer and cancer treatments on other organs and systems.
- Comprehensive physical examination, including gait, posture, and range of motion.
- Assessment of causative or contributing factors:[1]
 - Anemia.
 - Hypothyroidism.
 - Fluid/electrolyte imbalance.
 - Weight/caloric intake.
 - Sleep disturbances (e.g., insomnia, hypersomnia, sleep apnea, and restless legs syndrome).
 - Emotional disturbances (depression or anxiety), including psychiatric history and adversity during childhood.[11]
 - Pain.
 - Other treatment-related side effects (e.g., neuropathy or hot flashes).
 - Review of medication effects and effects caused by drug interactions (e.g., exacerbation of fatigue due to sedation or insomnia, worsening of depression, and cardiovascular effects).
 - Assessment of other comorbidities (e.g., alcohol and drug misuse and illicit substance use, cardiovascular or pulmonary diseases, endocrine dysfunction, neurological disorders, renal or hepatic dysfunction, infections, and gastrointestinal dysfunction).
 - Assessment of social, economic, and spiritual factors that can directly or indirectly exacerbate fatigue levels (by worsening emotional distress).
 - Assessment of functional status: physical activity levels and deconditioning.

An in-depth fatigue evaluation also includes an assessment of specific aspects of fatigue based on patient self-report:

- Onset.
- Duration.
- Pattern.
- Change in intensity and frequency over time.
- Exacerbating or alleviating factors.
- Associated patient distress.
- Interference with functioning.

Although there is no universally accepted standard for the measurement of fatigue, a variety of instruments can assess fatigue and related sequelae.[10,12-15][Level of evidence: II]; [16-19] These instruments range from single-item instruments screening tools to multi-item, multidimensional instruments used to conduct in-depth evaluations of fatigue. These instruments can be generally divided into three major categories:

- Very brief, single-item instruments that can be used for fatigue screening and longitudinal monitoring of fatigue (e.g., the VAS).
- Brief, multi-item but unidimensional instruments (e.g., the Brief Fatigue Inventory [BFI]).
- Comprehensive, multi-item, and multidimensional instruments (e.g., the Multidimensional Fatigue Inventory).

Table 2 delineates several instruments that are commonly used (in research and clinical practice) and have known psychometric properties. The use of a specific instrument in clinical practice is informed by what the instrument assesses and the objective of fatigue assessment at a specific time. For example, the VAS is used to screen for the presence or absence of fatigue in an expedited manner and to get a quick assessment of its severity. Multi-item but unidimensional assessments such as the BFI can be used to conduct an in-depth fatigue assessment, including fatigue severity, patterns, and impact on functioning. The multidimensional instruments can be used to conduct a comprehensive evaluation of fatigue in patients with complex fatigue patterns. These instruments assess fatigue severity, patterns, and impact on functioning, similar to the unidimensional assessments. Additionally, these instruments can be used in multiple fatigue domains (e.g., physical, affective, and cognitive).

Table 2. Commonly Used Fatigue Instruments

Instrument	Description	Reference
Single-item screening instruments		
Visual analog scale (VAS) for fatigue	Uses a 10-cm, 0- to 100-mm line; assesses severity only	Glaus, 1993 [3]
NCCN intensity tool	0-10 scale; assesses severity only	Mock et al., 2007 [2]
Multi-item, unidimensional instruments		
NCCN = National Comprehensive Cancer Network; QOL = quality of life.		

Instrument	Description	Reference
Brief Fatigue Inventory (BFI)	9 items; 0–10 numeric scale; measures severity of fatigue in past 24 h	Mendoza et al., 1999 [13]
Fatigue Symptom Inventory (FSI)	13 items; 0–10 numeric scale; measures fatigue severity, duration, and impact on QOL in past 7 d	Hann et al., 1998 [20]
Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-F)	13 items; 5-point, 0–4 scale; measures fatigue severity, duration, and impact on QOL in past 7 d	Yellen et al., 1997 [7]
Multi-item, multidimensional instruments		
Multidimensional Fatigue Inventory (MFI)	20 items; 7-point Likert scale; measures general, mental, and physical dimensions and activity level in past 24 h	Smets et al., 1995 [8]
FACIT-F	40 items; 5-point, 0–4 scale; includes 13-item FACIT-F scale; also measures physical, social/family, emotional, and functional dimensions	Yellen et al., 1997 [7]
European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire-Fatigue Module (EORTC QLQ-FA13)	13 items; measures physical, emotional, and cognitive aspects of fatigue, fatigue interference, and its sequelae	Weis et al., 2013 [9]
Revised Piper Fatigue Scale (PFS-R)	22 items; 11-point scale; measures behavioral, affective, sensory, and cognitive dimensions of fatigue	Piper et al., 1998 [10]
NCCN = National Comprehensive Cancer Network; QOL = quality of life.		

Proposed criteria for CRF are listed [below](#). These criteria have been adopted for inclusion in the *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision, Clinical Modification* (ICD-10-CM).[\[21\]](#)

Defining CRF as a diagnostic syndrome has potential advantages and disadvantages.[\[22\]](#) One of the possible advantages is that clinicians can document the presence or absence of fatigue in a reproducible fashion. The syndrome-based approach may also be useful in establishing appropriate reimbursement for the management of this finding. The potential disadvantage of this approach is that it may deter management of fatigue that does not reach the threshold of an ICD-10 diagnosis. The alternative to the syndrome-based approach (commonly used for depression) is a symptom-based approach, which is commonly used for phenomena such as pain and nausea. The utility of the following ICD-10 criteria for CRF has not been validated.

ICD-10 Criteria for CRF

CRF exists when the following symptoms have been present every day or nearly every day during the same 2-week period in the past month:

- A. Significant fatigue, diminished energy, or increased need to rest, disproportionate to any recent change in activity level, plus five or more of the following:
1. Complaints of generalized weakness or limb heaviness.
 2. Diminished concentration or attention.
 3. Decreased motivation or interest to engage in usual activities.
 4. Insomnia or hypersomnia.
 5. Experience of sleep as unrefreshing or nonrestorative.
 6. Perceived need to struggle to overcome inactivity.
 7. Marked emotional reactivity (e.g., sadness, frustration, or irritability) to feeling fatigued.
 8. Difficulty completing daily tasks attributed to feeling fatigued.
 9. Perceived problems with short-term memory.
 10. Postexertional fatigue lasting several hours.
- B. Clinically significant distress or impairment in social, occupational, or other important areas of functioning caused by the symptoms.
- C. Evidence from the history, physical examination, or laboratory findings that the symptoms are a consequence of cancer or cancer therapy.
- D. Symptoms not primarily a consequence of comorbid psychiatric disorders such as major depression, somatization disorder, somatoform disorder, or delirium.

As with other self-reported symptoms such as pain, it may be necessary to encourage the patient and other family members to report symptoms of fatigue to the medical staff. Information regarding the potential for fatigue due to the underlying disease or treatments, possible options for management, and the importance of reporting these symptoms is given to patients at the initiation of treatment.[23] Patients may not mention the fatigue they experience unless prompted by a health professional.

Several barriers hamper appropriate management of CRF. Some of these barriers were identified in phase 1 of an ongoing three-phase project related to the implementation of evidence-based guidelines for fatigue management from NCCN.[24] The most commonly identified barriers were the following:[24,25]

- The patient's belief that the physician would introduce the subject of fatigue if it were important (patient barrier).
- Lack of fatigue documentation (professional barrier).
- Lack of supportive care referrals (system barrier).

Evaluation of Anemia

The proper evaluation of anemia in cancer patients includes the following:

- A careful history and physical examination.
- An evaluation of the complete blood count and red blood cell indices.
- An examination of the peripheral blood smear.

In combination, the information from these investigations is often diagnostic.

One commonly used method for classifying anemia is to categorize the anemia by the size of the red blood cell, as measured by the mean corpuscular volume (MCV).

- Microcytic anemias are associated with an MCV of 79 fL or lower and include iron-deficiency anemia, thalassemia, and anemia of chronic disease.
- Macrocytic anemias are associated with an MCV higher than 101 fL and include anemias related to vitamin B₁₂ or folate deficiency, myelodysplasia, and liver disease.

Most anemias are normocytic, meaning that the MCV is in the normal range. This category includes the following:[26]

- Myelophthistic anemia (i.e., anemia due to neoplastic replacement of the bone marrow).
- Most chemotherapy-related anemias.
- Anemia due to renal or hepatic dysfunction.
- Hemolytic anemia.
- Aplastic anemia.

However, a mixed red blood cell population consisting of both microcytic and macrocytic cells (anisocytosis) may indicate a combined etiology, for example, chronic blood loss (microcytic) with resultant reticulocytosis (macrocytic). In this situation, the MCV may be in the normal range, but the red blood cell size distribution width would be elevated.

The peripheral blood smear examination, though often overlooked, remains an important step in the evaluation of anemia. For example, nucleated blood cells and teardrop-shaped red blood cells suggest myelophthistic anemia. Macro-ovalocytes and hypersegmented neutrophils often indicate megaloblastic anemia. Small target cells and basophilic stippling are associated with thalassemia.

Additional studies that are sometimes required to characterize anemia in a given patient include tests for the following:

- Vitamin B₁₂ or folate levels.
- Serum iron, transferrin, and ferritin levels.
- Erythropoietin level, the direct and indirect Coombs test, and/or examination of a bone marrow aspirate and biopsy.

In cancer patients, the underlying etiology is often multifactorial.

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Interventions

Much of the information regarding interventions for fatigue relates to healthy subjects, people in whom muscle fatigue is the primary etiology of the problem, or people in whom fatigue is secondary to treatment-related anemia.^[1,2][\[Level](#)

of evidence: II]; [3,4] Without a determination of the causative mechanisms of fatigue in oncology patients, interventions must be directed to symptom management and emotional support. Although some recommendations for the management of fatigue in oncology patients have been made, these are theoretical or anecdotal in nature and in general have not been the focus of scientific evaluation.

Published in 2013, a study conducted in patients with advanced cancer (N = 152) demonstrated that managing symptoms (e.g., pain, nausea, and decreased appetite) can have a significant positive impact on fatigue.[5] In this 12-week study, patients were randomly assigned to receive either monitoring and protocolized treatment of physical symptoms coordinated by a nurse or usual care (symptom management included in the standard oncologic care). Patients in the intervention group received tailored treatment for any of the identified troublesome symptoms. Fatigue levels, as measured by the Multidimensional Fatigue Inventory, showed significant improvement in the intervention group compared with the group receiving usual care. The intervention group also showed improvements in the following:[5]

- Specific fatigue dimensions.
- Interference by fatigue with daily life.
- Overall symptom burden.
- Symptoms of depression and anxiety.

Assessing patients for the appropriate target symptom for intervention is probably the most efficient way to help them improve their health-related quality of life and manage their fatigue symptoms.

Because the etiology and mechanisms regarding fatigue in cancer patients are varied, there is considerable need to personalize symptom management to provide goal-concordant care. Medical management is often directed at identifying specific and potentially reversible correlated symptoms, as in the following examples:

- Patients with fatigue and pain may benefit from titration of pain medications.
- Patients with fatigue and anemia may receive a transfusion of packed red blood cells (RBCs), nutritional interventions including iron-rich foods, supplemental iron or vitamins to correct an underlying deficiency, or injections of epoetin alfa.
- Patients with depressed mood and fatigue may be treated with antidepressants or psychostimulants.

Treatment of Anemia

Anemia in patients with cancer is best managed by treatment of the underlying cause. When the cause is obscure or there is no specific remedy, then treatment is supportive. Nutritional interventions, including the intake of nutrient-rich foods and supplements, are considered in addition to other treatment modalities.

The transfusion of packed RBCs is the most widely used and most rapid way to alleviate symptoms in cancer patients with symptomatic anemia. The likelihood of raising a patient's hemoglobin level is very high with transfusion, and the risks of complications are low. Nevertheless, repeated transfusions can be cumbersome, and the risk of blood-borne infection can be worrisome. Other risks include an acute transfusion reaction, transfusion-associated graft-versus-host disease, subtle immune modulation that occurs with transfusion, and iron overload in patients who receive repeated transfusions.[6]

The management of cancer-associated anemia using erythropoiesis-stimulating agents (ESAs) was established in the 2019 American Society of Clinical Oncology (ASCO)/American Society of Hematology (ASH) guidelines, which recommended the following:[7]

- ESAs (including biosimilars) may be offered to patients with chemotherapy-associated anemia whose cancer treatment is not curative in intent and whose hemoglobin levels have declined to lower than 10 g/dL. RBC

transfusion is also an option.

- With the exception of selected patients with myelodysplastic syndromes, ESAs should not be offered to most patients with non-chemotherapy-associated anemia.
- During ESA treatment, hemoglobin may be increased to the lowest concentration needed to avoid transfusions.
- Iron replacement may be used to improve hemoglobin response and reduce RBC transfusions for patients with or without iron deficiency who are receiving ESAs.
- ESAs increase the risk of thromboembolism, and clinicians should carefully weigh the risks of thromboembolism and use caution and clinical judgment when considering the use of these agents.[7]

Psychostimulants

Psychostimulants are a common pharmacological intervention for cancer-related fatigue (CRF); however, the evidence for their efficacy is mixed. Psychostimulants are drugs that interact with neurotransmitters and receptors in the brain to increase cortical function. Different types of psychostimulants work through various mechanisms to produce activity in the brain consistent with short-term improvement in energy level and psychomotor activity. These medications may also improve mood, attention, and concentration in some populations. Psychostimulant clinical trials for the management of fatigue include the following therapies (for information about levels of evidence and dosing used in the clinical trials, see Table 3):

- Methylphenidate.
- Dextroamphetamine.
- Modafinil.
- Armodafinil.

Psychostimulants are not approved by the U.S. Food and Drug Administration (FDA) for the treatment of CRF. However, preliminary evidence from randomized controlled studies [8-10] suggests that these medications might be helpful in a subpopulation of patients experiencing moderate to severe fatigue. Of the psychostimulants, methylphenidate is the most studied pharmacological agent for fatigue, yet the evidence for its efficacy is mixed.[11]

Table 3. Centrally Acting Stimulants for Adult Cancer Patients

Drug	Dosage	Outcome	Comments/Primary Side Effects
Dextroamphetamine (Dexedrine)	10 mg PO bid x8 d	No significant difference in fatigue.[12]	Schedule II. ^a Major potential interactions with citalopram and venlafaxine. Avoid in patients with uncontrolled hypertension, underlying CAD, and tachyarrhythmias.[13]
Methylphenidate (Ritalin)	5 mg PO bid titrated up to 20 mg tid over 6 wk	No difference in fatigue compared with placebo.[14] [Level of evidence: I]	Schedule II. ^a High-fat meals may increase AUC. Peak concentration 102 h postingestion. Do not use
AUC = area under the curve; bid = twice daily; CAD = coronary artery disease; CRF = cancer-related fatigue; LHRH = luteinizing hormone-releasing hormone; MAOI = monoamine oxidase inhibitor; PO = by mouth; prn = as needed; q2h = every 2 hours; qd = every day; SR = sustained release; SSRI = selective serotonin reuptake inhibitor; tid = three times daily. ^a As defined by the U.S. Controlled Substances Act.			

Drug	Dosage	Outcome	Comments/Primary Side Effects
	Titrate up to 54 mg/d PO (27 mg D-isomer) over 4 wk	Did not improve CRF compared with placebo.[15]	with MAOIs as it can precipitate hypertensive crisis. Antidepressants that increase norepinephrine can increase amphetamine side effects. Concomitant use with SSRI can increase SSRI concentrations. Avoid in patients with uncontrolled hypertension, underlying CAD, and tachyarrhythmias.[13]
	5 mg PO bid titrated every 3 d based on response	Dose-dependent improvement in CRF compared with placebo.[16]	
	5 mg PO q2h prn, up to 20 mg/d	CRF improved at day 15 with or without nursing telephone call.[17]	
	18 mg SR PO daily x2 wk	Did not improve CRF.[18]	
	10 mg PO qd	CRF improved at 10 w among men receiving LHRH agonists.[19]	
	5 mg PO bid x3 d each cycle	Did not improve CRF.[20]	
	5 mg PO prn titrated every 2–3 d to maximum 30 mg	CRF improved among men with prostate cancer.[9]	
Modafinil (Provigil)	200 mg PO qd	CRF improved in patients receiving chemotherapy with severe fatigue, but not mild/moderate fatigue.[10]	Schedule IV. ^a Avoid driving or operating machinery until effects are known. Do not take at bedtime. Peak concentration in 2–4 h. Food slows absorption by about 1 h but does not affect bioavailability. Decreases efficacy of birth control pills.
	100 mg PO bid (up to 400 mg/d) x6 wk	Did not improve CRF over placebo in patients with primary brain tumors.[21]	
	200 mg PO qd x15 d	Did not improve CRF over placebo in patients receiving docetaxel.[22]	
	100 mg PO qd on days 1–14 and 200 mg qd on days 15–28	Did not improve CRF over placebo in patients with lung cancer.[23]	
Armodafinil (Nuvigil)	50 mg PO bid	Did not improve daytime fatigue over placebo in cancer survivors with insomnia.[24]	Schedule IV. ^a Avoid driving or operating machinery until effects are known. Do not take at bedtime. Peak concentration in 2 h if fasting, slowed to as many
AUC = area under the curve; bid = twice daily; CAD = coronary artery disease; CRF = cancer-related fatigue; LHRH = luteinizing hormone-releasing hormone; MAOI = monoamine oxidase inhibitor; PO = by mouth; prn = as needed; q2h = every 2 hours; qd = every day; SR = sustained release; SSRI = selective serotonin reuptake inhibitor; tid = three times daily. ^aAs defined by the U.S. Controlled Substances Act.			

Drug	Dosage	Outcome	Comments/Primary Side Effects
	150 mg PO qd x56 d	Did not improve CRF over placebo in patients with multiple myeloma.[25]	as 4 h if fed; food does not affect bioavailability. Decreases efficacy of birth control pills.
AUC = area under the curve; bid = twice daily; CAD = coronary artery disease; CRF = cancer-related fatigue; LHRH = luteinizing hormone-releasing hormone; MAOI = monoamine oxidase inhibitor; PO = by mouth; prn = as needed; q2h = every 2 hours; qd = every day; SR = sustained release; SSRI = selective serotonin reuptake inhibitor; tid = three times daily. ^aAs defined by the U.S. Controlled Substances Act.			

Methylphenidate

Among all pharmacological treatment options for CRF, methylphenidate has the best evidence based on studies to date. The evidence for methylphenidate remains mixed, but it may play an important role in a subpopulation of cancer survivors.

One of the studies that demonstrated significant improvements over placebo for CRF used a mean dose of 27.7 mg of the D-isomer of methylphenidate as a study intervention.[8] The population that benefited was women who had completed chemotherapy for breast or ovarian cancer. The study design incorporated a titration to effect, so some patients who may have benefited may have received more than 27.7 mg of the drug. Furthermore, 11% of trial participants withdrew because of adverse events, compared with 1% in the placebo arm.

Conversely, an equally large randomized controlled trial assigned patients with early and advanced disease, who were either receiving treatment or not receiving treatment, to receive 54 mg of a long-acting methylphenidate preparation equaling 27 mg of the D-isomer or a placebo. This trial found no differences between the two groups in any of the fatigue outcomes.[15][[Level of evidence: I](#)] There were significant differences between groups for nervousness and appetite loss, with the methylphenidate arm scoring worse on both of those side effects.

Another large randomized placebo-controlled trial (n = 163) in patients with advanced cancer investigated immediate-release methylphenidate, starting at 5 mg twice daily.[14][[Level of evidence: I](#)] The dose was titrated up to 20 mg three times daily over a 6-week period. Improvement in fatigue and depression was nominally higher in the methylphenidate group than in the placebo group, but the differences did not reach clinically meaningful levels. There were no differences in adverse events between the two groups, including side effects such as anxiety, insomnia, loss of appetite, or heart palpitations.

Methylphenidate continues to show mixed results as a treatment for fatigue in patients with cancer. Several factors may cause this situation, including high placebo response, differential formulation and dosing, and heterogeneity of patient-related factors (e.g., early stage versus advanced stage of cancer, heterogeneity of cancer, cancer treatments, comorbidities, and concomitant medications). Based on the mixed trial results to date, methylphenidate may treat fatigue in specific subpopulations of cancer survivors. Immediate-release formulations with a shorter half-life appear to have a better safety and tolerability profile, with a lower risk of side effects, compared with longer-acting formulations.

Modafinil and armodafinil

The newer so-called wake-promoting agents, modafinil and armodafinil, are just beginning to be studied for CRF. Modafinil is a centrally acting, nonamphetamine central nervous system stimulant.[26] Armodafinil is the R-enantiomer of modafinil and an alpha-1 adrenoceptor agonist.[27] The FDA has approved modafinil and armodafinil for the treatment of narcolepsy, obstructive sleep apnea, and shift-work disorders but not for the treatment of CRF. These agents are also not indicated for use in children and adolescents.

The mechanism of action of modafinil and armodafinil is different from that of amphetamines, but the exact mechanisms by which these agents improve wakefulness are not known. On the basis of two promising open-label pilot trials,[28,29] a large randomized controlled trial evaluated modafinil for the treatment of CRF using 200 mg versus placebo in more than 850 patients who were receiving chemotherapy.[10] Patients had to have fatigue ratings of at least 2 out of 10 to be eligible for this study. During four cycles of chemotherapy, there were no significant differences between arms.

A randomized placebo-controlled trial (four-arm factorial study) comparing cognitive behavioral therapy (CBT) for insomnia (CBT-I) versus armodafinil (50 mg by mouth twice a day) found that CBT-I with and without armodafinil resulted in a clinically and statistically significant reduction of subjective daytime fatigue in cancer survivors with chronic insomnia.[24] Armodafinil alone did not show a statistically significant effect on fatigue for cancer survivors.

For both methylphenidate and modafinil, exploratory data have suggested that patients with more severe fatigue or more advanced disease may benefit from these drugs.[10,15] A small (N = 23), randomized, placebo-controlled study [9] using methylphenidate (titrated up to 30 mg/d) as an intervention failed to show statistical difference on the primary outcome measure, the Brief Fatigue Inventory (BFI) total score, or activity interference subscale. However, the methylphenidate group showed significant reductions in the BFI severity subscale scores compared with the reductions seen in the placebo group. The mean severity score at baseline was 6.5 for the methylphenidate group and 5.7 for the placebo group, placing these patients in a more severe fatigue category. A secondary analysis of the phase III trial that evaluated modafinil versus placebo for CRF also revealed that patients with more severe fatigue may have benefited from modafinil.[10] More research is needed to further evaluate whether psychostimulants are beneficial for patients experiencing more severe CRF.

Clinical considerations

The side effects most commonly described with the use of psychostimulants include the following:[8,10,15,30,31]

- Insomnia.
- Euphoria.
- Headache.
- Nausea.
- Anxiety.
- Mood lability.

High doses and long-term use may produce:

- Anorexia.
- Nightmares.
- Insomnia.
- Euphoria.
- Paranoia.
- Risk of cardiovascular complications.

Patients with cancer carry a higher risk of cardiovascular complications, depending on the type of cancer and cancer treatment (i.e., cardiotoxic chemotherapy regimens). Cardiovascular complications with psychostimulants can arise even in patients without any significant risk factors.[9] In the study using methylphenidate to treat CRF in patients with prostate cancer, 6 of 16 subjects (27%) in the methylphenidate group were withdrawn because of increased blood pressure and tachycardia. Notably, none of these subjects were being treated with known cardiotoxic chemotherapeutic regimens such as anthracyclines.[9]

Careful and continuous monitoring of certain cardiovascular parameters (mainly blood pressure and heart rate) is critical when psychostimulants are used to treat CRF. In certain complex cases, consultation with cardiology services may be considered. Cardiovascular issues are thought to be less of a risk with modafinil and armodafinil. The risk-benefit ratio may be considered, and patients may be evaluated for response and side effects, when these agents are used to treat CRF.

The package inserts for all Schedule IV stimulant medications (as defined by the U.S. Controlled Substances Act) carry boxed warnings that indicate the risk of abuse potential and/or risk of psychological dependence. In addition, boxed warnings for certain stimulant medications (methylphenidate and dextromethylphenidate products) indicate the risk of psychotic episodes.[30] Other stimulant medications (amphetamine, dextroamphetamine, lisdexamfetamine dimesylate, methamphetamine, and mixed salts of amphetamine products) carry boxed warnings alerting clinicians that misuse of these medications may cause serious cardiovascular adverse events, including sudden death.[32]

On the basis of limited clinical experience and a lack of evidence in randomized controlled trials, it might be reasonable to consider a psychostimulant such as methylphenidate or modafinil for the treatment of severe fatigue, particularly for short periods of time (a couple of weeks) in patients with advanced disease. When the use of these medications is being considered, clinicians should obtain informed consent, with a careful discussion of risks, benefits, and alternatives. Continuous monitoring of cardiovascular parameters is crucial when these medications are used, especially in patients with preexisting cardiovascular issues and in patients being treated with known cardiotoxic chemotherapeutic regimens (e.g., anthracyclines).

Longer-term psychostimulant therapy is not advisable at this time because there is limited information about its potential negative effects and longer-term benefits. Further research is needed in the form of CRF studies using psychostimulants in patients with depression or with drowsiness and sleep disturbance. In the design of these studies, it is important to consider patients with moderate to severe fatigue. These studies should also be performed for a longer period (>4 weeks) to account for the placebo wash-in period.[11]

Other Pharmacological Interventions

Bupropion

Bupropion is a stimulating antidepressant with a primarily dopaminergic and noradrenergic mechanism of action. Preliminary evidence from a small open-label study (N = 21) suggests that the sustained-release (SR) form of bupropion has potential as an effective therapeutic agent for treating CRF, with or without comorbid depressive symptoms.[33] [Level of evidence: II] Seizure, a rare but serious side effect of this agent, did not occur in this study (the maximum dose of bupropion SR used was 300 mg).

A small, double-blind, placebo-controlled trial of bupropion SR 150 mg daily versus placebo in a heterogeneous group of patients with cancer (N = 40) [34] demonstrated improvement in fatigue and quality of life as measured by the Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-F) scale ($P = .000$) at 4 weeks, compared with baseline assessments of symptoms. Adjustments for fatigue severity, depression, and cancer type did not modify the treatment effect on fatigue outcomes. No differences in adverse outcomes were noted between groups; however, the group receiving bupropion had a higher incidence of nausea and vomiting.[34]

Corticosteroids

Corticosteroids are by far one of the most commonly used medications for symptom control in patients with advanced cancer. These agents have potent anti-inflammatory effects and act by binding to cytoplasmic steroid hormone receptor and modulation of inflammatory gene transcription.[11] Dexamethasone is a potent anti-inflammatory agent that has been evaluated for the treatment of fatigue in patients with advanced cancer.[35] Eighty-four patients were randomly assigned to receive either dexamethasone 4 mg twice per day or a placebo for 14 days. The primary endpoint was improvement in fatigue from baseline to day 15, as measured by the FACIT-F scale. Investigators also evaluated depression, anxiety, and symptom distress. In the group who received dexamethasone, mean scores on the FACIT-F

scale were significantly improved by day 8 ($P = .005$) and at day 15 ($P = .008$). Physical well-being and physical distress were also significantly better in the group who received dexamethasone. Emotional scores and overall symptom distress were not significantly different. Adverse events, as measured by the Common Terminology Criteria for Adverse Events, version 3.0, did not differ between groups.

One limitation of this study was that it was only 2 weeks long, and longer-term use of dexamethasone is well known to be associated with unwanted side effects. Therefore, the risk versus benefit of treating fatigue with dexamethasone for more than 2 weeks requires investigation. Because fatigue has been associated with high levels of inflammation, this study is noteworthy in its evaluation of dexamethasone as an anti-inflammatory agent to alleviate fatigue.[35] The investigators did not assess inflammatory biomarkers; therefore, the proof of concept that modifying inflammation can reduce fatigue needs replication.

Dietary supplements

Dietary supplements comprise other, often popular, pharmacological interventions for CRF.

American ginseng

Ginseng, a popular supplement used to treat fatigue, has been evaluated in large, multisite clinical trials. On the basis of a promising phase II dose-finding study,[36] a phase III, randomized, placebo-controlled trial involved 364 patients with cancer who either were undergoing anticancer treatment or had completed treatment. Participants were randomly assigned to receive either 2,000 mg of American ginseng (specifically, Wisconsin ginseng) in the form of ground root in a capsule or a matching placebo. The primary endpoint was change in fatigue scores, as measured by the Multidimensional Fatigue Symptom Inventory-Short Form. At 4 weeks, the group receiving ginseng showed a trend toward significant improvement, while at 8 weeks, there was a significant and clinically meaningful difference favoring the ginseng group. There were no discernible side effects during the course of the trial, either within or between groups. [37-39]

Other supplements

Two additional supplements, coenzyme Q₁₀ and levocarnitine (L-carnitine), have been tested in large, randomized trials for the treatment of fatigue; however, they have failed to yield positive effects.

L-carnitine, a widely used dietary supplement, is believed to be helpful for the treatment of CRF because of its role in cellular energy metabolism and carnitine's ability to decrease pro-inflammatory cytokines. Promising pilot data led to the development and completion of a large (N = 376) phase III study in a multisite cooperative group setting.[40] Participants with moderate to severe fatigue were randomly assigned to receive either 10 g L-carnitine or a placebo for 4 weeks. The primary endpoint was change in average fatigue. Despite increases in mean levels of L-carnitine, there was not a statistically significant difference in fatigue between arms, with both arms reporting improvement during the study.[40]

Similarly, 300 mg of coenzyme Q₁₀ was tested against placebo in a double-blind randomized controlled trial of 236 breast cancer patients. Although supplementation led to sustained increases in plasma coenzyme Q₁₀, there were no significant differences between the groups over the 24-week study.[41] For more information, see [Coenzyme Q10](#).

Exercise

Studies suggest that exercise or physical activity has a beneficial effect on fatigue in patients during and after cancer treatment. The National Comprehensive Cancer Network (NCCN) guidelines [42] identify physical activity as an intervention for patients during and after treatment (category 1 intervention). Researchers have noted reductions in fatigue of about 35% and improvements in vitality of 30% in randomized trials.[43,44] Other documented benefits of exercise or physical activity include the following:

- Improved physical energy.

- Appetite stimulation.
- Improved memory.
- Enhanced functional capacity.
- Enhanced psychosocial well-being (improved outlook, sense of well-being, and quality of life).

Initial trials of exercise programs focused on women with breast cancer, but subsequent studies included men with prostate cancer and patients with multiple myeloma, lung cancer, nasopharyngeal cancer, non-Hodgkin lymphoma, colorectal cancer, and advanced cancers.[45-47]

Some studies had methodological weaknesses, including the following:[48][[Level of evidence: I](#)]; [49]

- Selection biases and nonrepresentative samples.
- Varied type (aerobic, anaerobic, or combined), dose (frequency and duration), and timing (during or after cancer treatment) of exercise prescription.
- Poor adherence to exercise interventions.
- Highly varied assessments of research variables and outcome measures.
- Lack of adequate control groups.

In a study of 545 breast cancer survivors who were, on average, 6 months postdiagnosis, increased physical activity was consistently related to both improved physical functioning and reduced fatigue and bodily pain. Prediagnosis physical activity was associated with better physical functioning at 39 months but generally unrelated to symptoms. Increased physical activity after cancer was related to less fatigue and pain and better physical functioning. Significant positive associations were found with moderate to vigorous recreational physical activity but not household activity. This study suggests that breast cancer survivors may be able to decrease fatigue and bodily pain and to better pursue daily activities by increasing their recreational physical activities after cancer.[50][[Level of evidence: II](#)]

A similar study of breast cancer survivors (N = 222) who were randomly assigned to a 3-month, multicomponent physical activity and behavior change intervention (Better Exercise Adherence after Treatment for Cancer [BEAT Cancer]) demonstrated reduced fatigue, depression, and anxiety symptomatology.[51]

A study of 1,033 patients with breast, gynecological, gastrointestinal, or lung cancer who had received chemotherapy in the past 2 weeks identified associations between level of exercise and fatigue and co-occurring sleep disturbance.[52] [[Level of evidence: II](#)] Three subgroups of patients were classified: no exercise, less exercise (<150 minutes per week), and recreational exercise (≥150 minutes per week). The most common reported exercise was walking. Compared with the other two groups, the no-exercise group had higher levels of morning fatigue, lower levels of morning and evening energy, and higher levels of sleep disturbance. Patients in the no-exercise group had fewer years of education, were more likely to be non-White, and had a higher body mass index and more comorbidities.[52][[Level of evidence: II](#)] These findings suggest associations among patient characteristics, engagement in exercise, and risk for fatigue.

Exercise for patients with advanced or terminal disease is difficult to study but may yield similar benefits. The ability of patients with advanced cancer who are in hospice care and on a physical therapy regimen to carry out activities of daily living reportedly improved in one study.[53][[Level of evidence: III](#)] Improved satisfaction with the physical therapy regimen was reported when family involvement in the program increased. A randomized study suggested that exercise improved fatigue during breast cancer treatment.[54][[Level of evidence: I](#)] An observational study of patients with advanced cancer found that fatigue was less severe in those who engaged in physical exercise.[55]

When educating patients about activity with respect to CRF, one important goal to consider is inclusion of 3 to 5 hours per week of moderate activity. It is critical that:

- Patients choose a type of exercise they enjoy.

- Providers discuss specific implementation strategies (type of exercise, time of day, days of the week, location of activity) to enable patients to make frequent activity a reality.

Beginning with lighter activity for shorter periods of time and building in intensity and length of time may be required. Studies have confirmed this can be safely done both during active treatment and after treatment is completed.[43]

Aerobic exercise

Two randomized controlled trials demonstrated the benefit of exercise in reducing fatigue during breast cancer treatment. A trial of a 12-week aerobic exercise program compared with usual care showed a nonsignificant improvement in fatigue 3 and 6 months later.[56][[Level of evidence: I](#)] Another trial that compared low-intensity and moderate- to high-intensity physical exercise with usual care showed that higher-intensity exercise (30 min/d, 5 d/wk) was beneficial in reducing fatigue.[57]

Limitations of both studies included the lack of a placebo control group and low participation rates. Low participation is a common finding in exercise studies of cancer patients, suggesting the need for tailored approaches to overcome barriers. The benefits shown in these studies are buttressed by a Cochrane review of 56 studies (including 4,068 participants), which concluded that aerobic exercise significantly reduced fatigue during or after cancer treatment.[58]

Anaerobic (resistance-training) exercise

Studies have also examined the use of resistance training to improve fatigue. In one large randomized controlled trial, 160 breast cancer patients (stages 0–III) were randomly assigned to a progressive resistance training intervention or a relaxation control intervention, twice weekly for 12 weeks. The primary endpoint was perceived fatigue, and the secondary endpoint was evaluated quality of life.[59]

Adherence to this group-based intervention program was as high as 97%. Significant improvements were noted between groups, favoring the resistance-training group for general fatigue ($P = .044$), especially for the physical fatigue subscale (mean difference = -0.8 ; 95% confidence interval, -1.5 to -0.2 , $P = .013$), but not for affective fatigue ($P = .91$) or cognitive fatigue ($P = .65$). For quality of life, significantly larger improvements regarding role function ($P = .035$) and pain ($P = .040$) were noted among exercisers compared with controls. This study demonstrated that resistance training was a feasible and efficacious strategy for improving fatigue and other components of quality of life.

Meta-analyses of aerobic, anaerobic, and combined exercise studies

Several literature reviews and meta-analyses have explored, with mixed results, the effect of exercise on fatigue. They have begun to examine which type of exercise—aerobic (cardio), anaerobic (resistance training), or a combination of the two—is most beneficial in ameliorating fatigue.

One large meta-analysis of breast cancer survivors identified 25 randomized controlled trials (including 3,418 patients) and examined the efficacy of exercise interventions for fatigue and physical functioning during and after treatment and at a 6-month follow-up.[60] Walking was noted to be the most prevalent exercise prescription among the studies reviewed. Improvements in physical functioning and fatigue were observed in the exercise studies during and after treatment, with slightly higher improvements in patients who received the intervention posttreatment. Although combined aerobic and anaerobic groups demonstrated slightly more improvement in physical functioning compared with controls, there were not significant differences in physical functioning and fatigue when all three groups were compared.

A 2018 systematic review and meta-analysis identified 245 studies of all cancer types, explored nonpharmaceutical interventions for fatigue during and after treatment, and conducted an indirect-comparisons meta-analysis among intervention types.[61] In this analysis, aerobic and anaerobic exercise improved fatigue more than usual care, with moderate to large effect sizes noted (standardized mean difference [SMD] for aerobic exercise, -0.53 ; 95% credible interval [CrI], -0.80 to -0.26 ; SMD for anaerobic exercise, -0.53 ; 95% CrI, -1.02 to -0.03). However, combined aerobic and anaerobic exercise demonstrated the most improvement, with a large effect size (SMD, -0.67 ; 95% CrI, -1.01 to -0.34).

A study of pooled baseline data from three studies ($n = 436$) investigated reallocating sedentary time to an equal amount of light or moderate-to-vigorous engagement in physical activity in patients with breast cancer. The study showed that fatigue improved in these patients.[62][[Level of evidence: III](#)] Specifically, 30 minutes of reallocated light activity improved motivation and engagement in activity ($\beta = -0.21$). Engagement in moderate-to-vigorous activity reduced general fatigue ($\beta = -0.34$) and physical fatigue ($\beta = -0.47$) and improved activity ($\beta = -0.48$). However, quality of life did not improve for any level of engagement in physical activity, as highlighted in other studies. Evidence suggests that reallocating sedentary behavior to engagement in physical activity can improve fatigue.

Limitations remain regarding the need to identify a more exacting exercise prescription, including the need to identify the type, intensity, frequency, and resting intervals to fully incorporate into cancer practice and survivorship care plans.[47]

Other exercise modalities

Variations of exercises that have a mind-body component are being studied for their effects on CRF; popular interventions include complementary modalities such as [yoga](#), [qigong](#), and [tai chi](#).[\[63-65\]](#) These modalities are unique in that they incorporate cognitive and spiritual elements with movement, stretching, and balance.

Yoga

Yoga is an ancient system of practices used to balance the mind and body through exercise, meditation (focusing thoughts), and control of breathing and emotions. Yoga has been shown to improve fatigue in cancer survivors in several pilot and larger randomized controlled trials (NCCN category 1 intervention).[\[66\]](#)

In one pilot study, 12 weeks of yoga was compared with a health education intervention in a control group in 31 breast cancer survivors.[\[67\]](#) The primary outcome was change in fatigue measured at baseline, immediately posttreatment, and 3 months after completion of treatment. Fatigue severity declined significantly from baseline to posttreatment and over a 3-month follow-up in the yoga group, relative to controls ($P = .032$). In addition, the yoga group had significant increases in vigor relative to controls ($P = .011$).

Similarly, in a larger randomized controlled trial, investigators examined the effect of two 90-minute hatha yoga sessions per week for 12 weeks delivered in a group setting, compared with a wait-list control in 181 breast cancer survivors.[\[68\]](#) Fatigue and vitality immediately posttreatment and at 3 months posttreatment were the endpoints of the study. Investigators noted significant improvement in fatigue at 3 months posttreatment, as well as improved vitality immediately posttreatment and at 3 months posttreatment. However, the study failed to find significant differences in fatigue immediately posttreatment.

In one large, multicenter phase III randomized controlled trial, investigators examined the effect of a standardized 4-week yoga therapy program (Yoga for Cancer Survivors [YOCAS]) on fatigue compared with standard survivorship care in 410 cancer survivors.[\[66\]](#) Compared with participants receiving standard survivorship care, the YOCAS participants demonstrated significantly greater improvements in fatigue ($P < .01$), as well as decreased interference of fatigue in walking, physical activity, and quality of life (all $P < .05$). Improvements in fatigue resulting from yoga accounted for significant proportions of the improvements in walking (44%), physical activity (53%), and quality of life (45%; all $P < .05$). Improvements in overall sleep quality and reductions in daytime dysfunction (e.g., excessive napping) resulting from yoga significantly mediated the effect of yoga on fatigue (22% and 37%, respectively, both $P < .01$).

A meta-analysis (including 10 studies of cancer survivors) examining yoga for fatigue found that yoga demonstrated a significant improvement in fatigue over usual care, with a moderate effect size (SMD, -0.68 ; 95% CrI, -0.93 to -0.43).[\[61\]](#)

The limitations of these studies include study designs that varied in the type of yoga and its duration, frequency, and number of weeks; failed to include attention control comparisons; and varied in fatigue assessment measures.

Qigong

Qigong is a traditional Chinese mind/body exercise and meditation that uses slow and precise body movements with controlled breathing and mental focusing to improve balance, flexibility, muscle strength, and overall health. One fairly large study evaluated medical qigong for CRF in a heterogeneous group of 162 patients during or after cancer treatment.[63] This study reported significant improvements in fatigue and several other aspects of quality of life for the intervention group versus usual care.

The qigong intervention was delivered in 90-minute group sessions, twice a week for 10 weeks, for 1,800 minutes of treatment. The usual-care group did not receive any group meetings or additional provider interaction. It is therefore difficult to say what qigong uniquely provided over and above nonspecific or group-interaction effects. It is also not known how much survivors would need to continue performing qigong to maintain benefits. There were no adverse events in this study, so other than time and resource expenditure, it is difficult to pinpoint a downside to encouraging patients to adopt such an activity. One important strength of the study was the collection of serum to measure markers of inflammation. At the end of 10 weeks, the C-reactive protein level of patients in the medical qigong group decreased by 3.6 mg/L, while patients in the usual-care group experienced an increase of 19.57 mg/L, a statistically significant difference.[63]

A second smaller study (N = 96) that compared a qigong group to a wait-list control group evaluated fatigue using the BFI as a secondary outcome; it also assessed a biological measure, salivary cortisol.[64] This study did not find any significant difference in fatigue or cortisol between groups. The intervention dose in this study, comprising five 40-minute sessions over 6 weeks of radiation therapy in women diagnosed with breast cancer, was much lower than the intervention dose in the larger study described above.

The major weakness limiting interpretation and integration of both of these studies, despite differing results, is that there was no attempt to control for attention or any of the social aspects of the intervention.

In a third small study (N = 76), men with prostate cancer undergoing radiation therapy were randomly assigned to qigong/[tai chi](#), light exercise, or a waiting list.[65] The qigong/[tai chi](#) group reported improvements in sleep duration midway during radiation therapy treatment (6.7 hours vs. 7 hours); however, this effect was not durable. There were no differences between groups in fatigue or sleep outcomes, suggesting that this may not be an effective intervention during radiation therapy for prostate cancer. The symptoms of fatigue and poor sleep were highly correlated with the physical symptom burden of men with prostate cancer.[65]

Tai chi

Tai chi is a Chinese martial arts activity that involves deep breathing, exercise, and slow movement with a meditative aspect, connecting the individual's physical, mental, and emotional states. Tai chi has been examined for its effect on cancer symptoms, including CRF.

Investigators conducted a randomized controlled trial to compare the effect of tai chi versus low-impact exercise on CRF during treatment for 91 lung cancer patients.[69] Tai chi sessions were conducted every other day for 12 weeks during each course of chemotherapy across four courses of treatment. Study assessments were conducted before the first and third courses of chemotherapy and at the end of the fourth course. Fatigue scores increased in all patients. However, in the tai chi group at 6 weeks, general and physical fatigue subscale scores were lower ($P < .05$) and vigor subscale scores were higher, compared with the scores of the exercise group ($P < .05$). These scores were also better in the tai chi group at 12 weeks ($P < .05$). No other differences existed between groups.[69]

In a subsequent meta-analysis, including six studies and more than 370 cancer patients, researchers noted significant and positive improvement in short-term CRF in patients with breast and lung cancer, but not in patients with prostate cancer.[70] A longer intervention period (>8 weeks) demonstrated greater improvements in CRF, and these effects were noted to be superior to the effects of physical exercise and psychological support. However, the effects of tai chi on long-term CRF remain unclear.

CBT

CBT has long been used to treat a variety of psycho-physiological problems, with therapy focusing on the thoughts (cognition) and functional behaviors relevant to the presenting problems. While most of the CBT research for CRF has focused on the survivor period, CBT and CBT variants (e.g., CBT-I and mindfulness-based cognitive therapy) have been shown to be useful during both active treatment and the survivor period.[46]

In the context of active treatment (e.g., chemotherapy, radiation therapy, surgery), CBT plus hypnosis may be effective for patients struggling with CRF.[71] Significant decreases in fatigue were reported over a 6-week course of psychotherapy during radiation therapy, compared with a control group. At a 6-month follow-up, the CBT group continued to experience significantly improved fatigue, compared with the control group.

In a randomized clinical trial, 98 mixed-type cancer survivors (intervention group = 50, wait-list control = 48) experiencing severe fatigue not attributable to a specific somatic cause were provided individual CBT.[72][[Level of evidence: I](#)] The CBT focused on each participant's unique pattern of the following six possible factors that might perpetuate their post-cancer treatment fatigue:

- Insufficient coping with the experience of cancer.
- Fear of disease recurrence.
- Dysfunctional cognition regarding fatigue.
- Dysregulation of sleep.
- Dysregulation of activity.
- Low social support/negative social interactions.

The number of therapy sessions varied according to the number of perpetuating factors (range, 5–26 one-hour sessions; mean: 12.5 sessions). Results showed a clinically significant decrease in fatigue severity and functional impairment.

Fatigue-related improvements that occur during CBT for CRF can be maintained for extremely long periods of time, even without periodic, long-term follow-up (booster) sessions that are usually a component of CBT. In a 10-year follow-up of 81 individuals who completed a CRF CBT protocol,[73] fatigue levels increased among cancer survivors over the 10-year follow-up period, compared with the initial post-CBT assessment. In addition, at the 10-year follow-up, fatigue levels continued to be higher among cancer survivors compared with general-population controls. However, more than half of the cancer survivors (52%) who recovered from severe fatigue at the time of the post-CBT assessment maintained their low fatigue levels at the 10-year follow-up. While levels of fatigue deteriorated over time, the strong maintenance gains for more than half of the study population suggest that CBT for CRF can help control fatigue over long periods of time.

While CBT and medication therapy may often work hand in hand, some studies show that CBT alone has a more powerful impact on fatigue than medication alone. In a 7-week, double-blind treatment study,[24] 96 cancer survivors with cancer-related insomnia and fatigue were randomly divided into four groups:

- CBT-I plus placebo (twice a day).
- CBT-I plus armodafinil (50 mg twice a day).
- Placebo alone (twice a day).
- Armodafinil alone (50 mg twice a day).

The study found a significant reduction of fatigue with CBT alone or with CBT plus armodafinil (though the drug provided limited additive benefit compared with CBT alone), and no improvement with armodafinil alone.[24] Furthermore, the armodafinil group showed significantly less improvement than did the placebo-alone group.

Patient Education

Informing patients about the risk of fatigue and educating them about strategies to reduce fatigue are valuable adjuncts to other management strategies. However, a Cochrane review of educational interventions for CRF in adults cautions that educational interventions should be part of a more-comprehensive approach to managing fatigue.[74]

Specific techniques for the management of fatigue include the following:

- Differentiation of fatigue from depression.
- Assessment for presence of correctable correlates or causes of fatigue (e.g., dehydration, electrolyte imbalance, dyspnea, anemia).
- Evaluation of patterns of rest and activity during the day as well as over time.
- Determination of the level of attention fatigue and encouragement of attention-restoring activities (e.g., walking, gardening, bird watching).
- Providing anticipatory guidance regarding the likelihood of experiencing fatigue and the fatigue patterns associated with particular treatments.
- Encouragement of activity/planned exercise programs within individual limitations; making goals realistic by keeping in mind the state of disease and treatment regimens.
- Education of individuals and families about fatigue related to cancer and its treatment.
- Helping people with cancer and their families identify fatigue-promoting activities and develop specific strategies to modify these activities.
- Suggesting individualized environmental or activity changes that may offset fatigue.
- Maintaining adequate hydration and nutrition.
- Recommending physical therapy referral for people with specific neuromusculoskeletal deficits.
- Recommending respiratory therapy referral for people with dyspnea that is a contributing factor to fatigue.
- Scheduling important daily activities during times of least fatigue and eliminating nonessential, stress-producing activities.
- Addressing the negative impact of psychological and social stressors and how to avoid or reduce them.
- Evaluating the efficacy of fatigue interventions on a regular and systematic basis.

In a controlled trial of patients who reported the symptom cluster of pain and fatigue while receiving chemotherapy, a nursing behavioral intervention produced improvements in quality of life and decreased symptom burden relative to usual care.[75,76][[Level of evidence: I](#)] These intriguing results need to be further explored in patient populations other than women with breast or gynecological malignancies.

As researchers and practitioners have learned with pain, misconceptions and a lack of knowledge may prove to be patient- and provider-related barriers to successful assessment and management. A quasi-experimental study tested a multisystem educational approach to improving both pain and fatigue management.[77] The approach consisted of the following:

- Education and assessment of patients regarding the management of pain and fatigue, with phone calls every 2 weeks for 3 months.
- Education of providers about pain and fatigue assessment and management, including monthly newsletters.
- An effort to engage with an internal advisory board.
- Efforts aimed toward research nurses to refer earlier to supportive care services.

Over a 3-month period, the educational intervention resulted in increases in knowledge and a decrease in barriers related to management of pain and fatigue. Of note, important patient barriers related to fatigue management included

the following beliefs:[77][[Level of evidence: II](#)]

- Fatigue is inevitable.
- Fatigue can indicate worsening of disease.
- Treating the cancer is more important than treating fatigue.
- Reporting fatigue will cause a patient to be perceived as a complainer.

Providing patient education about strategies to reduce fatigue may help eliminate the barriers related to managing fatigue.

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Latest Updates to This Summary (11/06/2024)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Interventions

Updated [Table 3](#), Centrally Acting Stimulants for Adult Cancer Patients, to add new information about methylphenidate to treat cancer-related fatigue (cited Stone et al. as reference 14 and level of evidence I).

The [Methylphenidate](#) subsection was extensively revised.

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About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the pathophysiology and treatment of fatigue. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

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- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

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The lead reviewers for Fatigue are:

- Katharine Brock, MD, MS (Children's Healthcare of Atlanta)
- Marilyn J. Hammer, PhD, DC, RN, FAAN (Dana-Farber Cancer Institute)
- Jayesh Kamath, MD, PhD (University of Connecticut Health Center)
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Updated: November 6, 2024

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